

Neural Network Approximation of Dynamical Systems

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Abstract

Neural networks have become a promising data-driven tool for modelling physical systems, offering an alternative approach for approximating dynamical behaviour without explicitly solving the governing equations. This work investigates the ability of a multilayer perceptron (MLP) to approximate the temporal evolution of dynamical systems with increasing complexity. Six representative systems were considered: the harmonic oscillator, damped harmonic oscillator, forced harmonic oscillator, Duffing oscillator, simple pendulum, and double pendulum.

A common neural network architecture was trained using time as the only input variable, while the corresponding physical state was used as the output. Parameter sweep experiments were performed for each system to evaluate how variations in physical parameters influence prediction accuracy. Model performance was assessed using the mean squared error (MSE), mean absolute error (MAE), maximum error, and coefficient of determination (R^2).

The results show that the MLP can accurately approximate the investigated systems across a wide range of explored conditions. The experiments indicate that prediction accuracy is influenced more strongly by the temporal complexity of the generated trajectories than by the presence of nonlinear terms in the governing equations alone. Higher oscillation frequencies, longer prediction intervals, and sensitivity to initial conditions were found to increase prediction errors.

These findings demonstrate the capability of simple feedforward neural networks for approximating deterministic dynamical systems and provide insight into the factors that limit their predictive accuracy.

1 Introduction

The modelling of physical systems has traditionally relied on mathematical formulations derived from fundamental laws, followed by numerical methods

to solve the resulting differential equations [17]. These approaches have been highly successful for a wide range of systems; however, obtaining accurate solutions can become increasingly challenging when the governing dynamics are nonlinear, high-dimensional, or sensitive to initial conditions [17]. In recent years, machine learning techniques have emerged as an alternative approach, allowing complex relationships between variables to be approximated directly from data [1].

Artificial neural networks have demonstrated significant potential in scientific computing due to their ability to approximate nonlinear functions [9]. The universal approximation theorem states that feedforward neural networks can represent continuous functions with arbitrary accuracy given sufficient model capacity [9]. However, this theoretical property does not guarantee efficient training or accurate generalization, particularly when the underlying dynamics exhibit complex temporal behaviour [7]. Nevertheless, neural networks provide a promising framework for learning the evolution of dynamical systems directly from generated data.

The performance of neural networks depends strongly on the characteristics of the dynamics being approximated. Linear systems generally produce smooth and predictable trajectories, whereas nonlinear systems can introduce phenomena such as amplitude-dependent frequencies, complex oscillations, and chaotic behaviour [17]. Systems exhibiting sensitivity to initial conditions present an additional challenge, since small variations in the initial state can lead to significantly different trajectories over time [13]. Understanding how these properties influence prediction accuracy is essential for evaluating the limitations of data-driven approaches in physical modelling.

In this work, the capability of a multilayer perceptron (MLP) to approximate several deterministic dynamical systems with increasing levels of complexity is investigated. Six systems are analysed: the harmonic oscillator, damped harmonic oscillator, forced harmonic oscillator, Duffing oscillator, simple pendulum, and double pendulum. A common neural network architecture is used throughout the experiments, and systematic parameter sweeps are performed to study how variations in physical parameters influence prediction performance.

The main objective of this work is to determine whether prediction difficulty is primarily related to the mathematical nature of the governing equations or to the complexity of the resulting trajectories. By comparing linear, nonlinear, and chaotic systems under a consistent framework, this study provides insight into the capabilities and limitations of simple feedforward neural networks for approximating deterministic physical dynamics.

2 Methodology

2.1 Neural Network Structure

The dynamical systems investigated in this work were approximated using a feedforward neural network implemented in PyTorch. The objective of the neural network was to learn the relationship between time and the corresponding physical state of each system, without having explicit information about the differential equations governing these systems during training.

For systems with a single state variable, the network receives time (t) as input and predicts the corresponding position:

$$x(t) = NN(t) \quad (1)$$

where $x(t)$ represents the approximation of the physical solution obtained by the neural network.

For the double pendulum system, however, the output dimension is larger since the network predicts two angular positions:

$$(\theta_1(t), \theta_2(t)) = NN(t) \quad (2)$$

The general architecture, although modified for some specific experiments, consists of three fully connected layers with 128 neurons each. The activation function used between layers is the hyperbolic tangent function (Tanh), allowing the network to represent the nonlinear relationships present in the physical systems studied.

Additionally, the network parameters were optimized using the Adam optimizer [10] with a learning rate of 5×10^{-4} . The training process was performed for 500 epochs using Mean Squared Error (MSE) as the loss function:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3)$$

where y_i represents the reference solution generated by the numerical simulation, and $\hat{y}_i$ represents the prediction produced by the neural network.

The dataset was divided into training and testing subsets using an 80/20 split. To ensure reproducibility, a fixed random seed was used throughout all experiments, unless otherwise indicated.

2.2 Dataset Generation

The datasets used in this work were generated through numerical simulations of different dynamical systems. Each system was simulated independently,

producing time-dependent trajectories that were later used as training and evaluation data for the neural network model.

For each system, the input variable was the time coordinate (t), while the target variables corresponded to the physical state of the system, either position or angular displacement.

The generated datasets consisted of trajectories sampled uniformly over a fixed simulation interval. Although the simulation duration and number of samples varied between experiments, each trajectory generally contained between 1,000 and 3,000 samples, allowing the neural network to learn the temporal evolution of the physical process.

2.2.1 Harmonic Oscillator

The harmonic oscillator was used as the simplest dynamical system investigated in this study. Its motion is described by the differential equation:

$$\ddot{x} + \omega^2 x = 0 \quad (4)$$

where x represents the displacement and ω represents the angular frequency.

The analytical solution of this equation is:

$$x(t) = A \cos(\omega t + \phi) \quad (5)$$

where A is the amplitude and ϕ is the initial phase. The generated dataset consists of pairs $(t, x(t))$, where the neural network was trained to approximate the relationship between time and displacement.

2.2.2 Damped Harmonic Oscillator

To introduce dissipative effects, a damping term was added to the harmonic oscillator equation:

$$\ddot{x} + \beta \dot{x} + \omega^2 x = 0 \quad (6)$$

where β represents the damping coefficient. Unlike the ideal harmonic oscillator, the amplitude decreases over time due to energy dissipation. The generated trajectories were obtained by numerically solving the differential equation and storing the resulting displacement evolution.

The dataset consisted of time-position pairs, allowing the neural network to learn the effect of damping on the behaviour of the physical system.

2.2.3 Forced Harmonic Oscillator

The forced harmonic oscillator extends the damped harmonic oscillator by introducing a periodic external force. This excitation continuously supplies energy to the system, resulting in more complex dynamical behaviour that depends on both the natural frequency of the oscillator and the frequency of the external force.

The motion of the system is governed by:

$$\ddot{x} + \beta\dot{x} + \omega_0^2x = F_0 \sin(\Omega t) \quad (7)$$

where F_0 represents the forcing amplitude and Ω represents the forcing angular frequency. The parameters β and ω_0 correspond to the damping coefficient and the natural angular frequency, respectively.

Unlike the previous systems, the response of the forced oscillator is determined not only by its intrinsic properties but also by the characteristics of the external excitation. Depending on the relationship between the natural and forcing frequencies, the system may exhibit resonance [12], where the oscillation amplitude increases significantly.

As in previous experiments, the trajectories were generated by numerically solving the governing differential equation over a fixed simulation interval. The resulting dataset consisted of pairs $(t, x(t))$, where the input corresponds to the time coordinate and the output represents the corresponding displacement.

Several experiments were performed by varying the forcing amplitude and forcing frequency while keeping the remaining physical parameters constant. This allowed the influence of external excitations on the neural network's ability to approximate the physical system to be evaluated.

2.2.4 Duffing Oscillator

The Duffing oscillator was introduced to evaluate the performance of the neural network on a nonlinear dynamical system. Unlike harmonic oscillators, where the restoring force is proportional to the displacement, the Duffing oscillator includes a nonlinear restoring term that can produce complex behaviors depending on the system parameters [17].

The governing equation of the Duffing oscillator is

$$\ddot{x} + \delta\dot{x} + \alpha x + \beta x^3 = \gamma \cos(\Omega t), \quad (8)$$

where δ represents the damping coefficient, α and β control the linear and nonlinear stiffness terms, γ is the forcing amplitude, and Ω is the forcing

angular frequency. The nonlinear term (βx^3) introduces a dependence between the restoring force and the amplitude of the oscillation. Depending on the chosen parameters, the system can exhibit different dynamic regimes, including periodic oscillations, amplitude-dependent frequencies, and more complex responses.

The trajectories were generated by numerically integrating the Duffing differential equation for a predefined set of initial conditions and physical parameters. Each simulation produced a time-dependent displacement trajectory ($x(t)$), which was used to construct the dataset. The neural network received the time coordinate t as input and learned to approximate the corresponding system state.

To analyze the influence of the nonlinear dynamics, experiments were performed by varying selected Duffing parameters while maintaining the remaining parameters fixed. In particular, the damping coefficient (δ) was modified to study how changes in energy dissipation affected the approximation accuracy of the neural network.

The Duffing oscillator represents a transition between classical oscillatory systems and more complex nonlinear dynamics, providing a challenging benchmark for evaluating whether neural networks can learn physical behaviors beyond simple analytical solutions.

2.2.5 Simple Pendulum

The simple pendulum was used as a nonlinear mechanical system to evaluate the ability of the neural network to approximate the dynamics of a physical system with angular motion. Unlike harmonic oscillators, the restoring force of a pendulum depends on the sine of the angular displacement, introducing nonlinear behavior for large oscillations.

The equation of motion of the pendulum is given by

$$\ddot{\theta} + \frac{g}{L} \sin(\theta) = 0, \quad (9)$$

where θ represents the angular displacement, g is the gravitational acceleration, and L is the pendulum length. For small angular displacements, the approximation ($\sin(\theta) \approx \theta$) transforms the system into a harmonic oscillator [6]. However, in this work the complete nonlinear equation was used, allowing the neural network to learn the full pendulum dynamics without simplifying assumptions.

The trajectories were generated by numerically solving the equation of motion for different initial conditions and physical parameters. Each

simulation produced the evolution of the angular position ($\theta(t)$), which was used as the target variable of the neural network. Therefore, the generated dataset consisted of pairs $(t, \theta(t))$, where the input was the time coordinate and the output was the corresponding angular displacement.

Several experiments were performed by varying the gravitational acceleration, pendulum length, and initial angular conditions. These parameter sweeps allowed the analysis of how changes in the physical configuration affected the difficulty of the learning problem and the approximation accuracy of the neural network.

The simple pendulum provides an intermediate benchmark between analytically solvable oscillatory systems and more complex nonlinear systems, as it introduces nonlinear dynamics while maintaining a single degree of freedom.

2.2.6 Double Pendulum

The double pendulum was selected as the most complex dynamical system in this study. Unlike the previous systems, which contain a single state variable, the double pendulum is a coupled system with two interacting degrees of freedom. The motion of each pendulum depends not only on its own state but also on the position and velocity of the other pendulum.

The system consists of two pendulum masses connected sequentially, with angular coordinates θ_1 and θ_2 . Its dynamics are governed by a coupled set of nonlinear differential equations derived from Lagrangian mechanics [6]. These equations depend on the masses m_1 and m_2 , the lengths L_1 and L_2 , the gravitational acceleration g , and the initial conditions of the system.

The state of the system can be represented as

$$y(t) = [\theta_1(t), \theta_2(t)], \quad (10)$$

where θ_1 and θ_2 describe the angular positions of the first and second pendulum links, respectively.

The trajectories were generated through numerical simulation of the double pendulum equations of motion. For each simulation, the resulting angular trajectories were sampled over a fixed time interval, producing datasets composed of time-state pairs:

$$(t, [\theta_1(t), \theta_2(t)]). \quad (11)$$

In contrast with previous experiments, the neural network was required to approximate a multidimensional output, predicting both angular coordinates simultaneously.

Due to the nonlinear coupling between the two pendulums, small variations in the initial conditions can lead to significantly different trajectories over time. This sensitivity to initial conditions is a characteristic feature of chaotic systems [17] and makes the double pendulum a challenging benchmark for data-driven modeling approaches.

Experiments were performed by modifying physical parameters such as pendulum lengths and initial angular conditions. These parameter variations were used to evaluate how changes in the underlying dynamics affected the neural network’s approximation capabilities.

The double pendulum represents the highest level of complexity considered in this work, extending the evaluation from simple periodic systems to nonlinear and chaotic dynamics.

3 Results

3.1 Harmonic Oscillator

The harmonic oscillator was used to evaluate the performance of different neural network architectures before applying them to more complex and demanding physical systems. Instead of investigating the influence of physical parameters, these experiments focused on evaluating different neural configurations by varying the number of hidden layers, the number of neurons per layer, and the learning rate, while keeping the physical system fixed.

Exp	Learning Rate	Hidden Layers	Neurons per Layer	MSE	MAE	Maximum Error
1	0.001	3	64	0.000869	0.021611	0.075934
2	0.001	3	128	0.000344	0.014924	0.051416
3	0.001	3	256	0.016496	0.085416	0.334049
4	0.0005	3	256	0.000046	0.005407	0.033024
5	0.0005	3	128	0.000105	0.006760	0.045358
6	0.0005	4	64	0.000110	0.008516	0.027999
7	0.0005	4	128	0.001400	0.029866	0.088467
8	0.001	4	128	0.000787	0.020596	0.087798

Table 1: Neural network configurations evaluated for the harmonic oscillator.

Table 1 summarises the performance obtained by the different neural network architectures tested. In general, the network achieved low prediction errors across the different configurations, demonstrating that even relatively

simple fully connected architectures are capable of accurately learning the sinusoidal dynamics of the harmonic oscillator.

Increasing the network size did not always improve performance. For example, increasing the number of neurons per layer from 128 to 256 while maintaining the learning rate at 10^{-3} increased the MSE from 3.44×10^{-4} to 1.65×10^{-2} , indicating that larger models can be more difficult to optimise under the same training conditions. However, reducing the learning rate to 5×10^{-4} improved the performance, reducing the MSE to 4.6×10^{-5} in experiment 4, which corresponds to the best performance achieved in these experiments.

Similarly, increasing the depth of the network did not consistently improve the results. Although some deeper architectures achieved competitive performance, they generally produced higher errors than the best-performing structure with three hidden layers. These observations suggest that, for a simple linear dynamical system such as the harmonic oscillator, increasing model complexity provides limited benefits once sufficient representational capacity has been reached.

Based on these experiments, the configuration consisting of three hidden layers with 128 neurons per layer, Tanh activation, and a learning rate of 5×10^{-4} was selected as the baseline architecture for the remaining experiments. Although some configurations achieved slightly lower errors in individual cases, this model was chosen because it provided a consistent balance between prediction accuracy, training stability, and computational efficiency. The objective of this selection was not to identify the absolute optimal architecture, but rather to establish a reliable reference model for comparing the influence of different physical parameters on the learning process. This configuration achieved an MSE of 1.05×10^{-4} , an MAE of 6.76×10^{-3} , and a maximum prediction error of 4.54×10^{-2} .

3.2 Damped Harmonic Oscillator

Following the validation of the neural network architecture using the harmonic oscillator, the damped harmonic oscillator was introduced to evaluate the model’s ability to learn dissipative dynamics. Unlike the ideal harmonic oscillator, where the oscillation amplitude remains constant, the presence of the damping coefficient (β) introduces an exponential decay in the oscillation amplitude, increasing the complexity of the underlying temporal pattern.

A total of 50 experiments were performed, evaluating ten different damping coefficients ($\beta = 0.1$ to 1) and repeating each configuration with five different random seeds. The neural network architecture and training param-

ters were kept fixed throughout all experiments according to the configuration selected in Section 3.1.

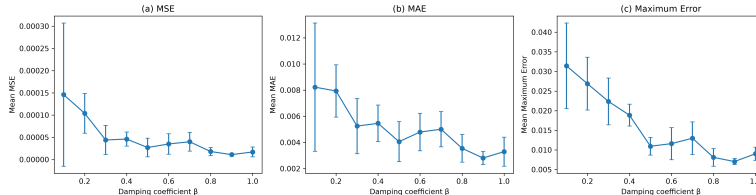


Figure 1: Prediction error metrics for the damped harmonic oscillator across different damping coefficients. The figure shows the mean squared error (MSE), mean absolute error (MAE), and maximum prediction error obtained for each value of the damping coefficient.

The obtained results are summarised in Figure 1, where the mean and standard deviation are shown for each damping coefficient. For all evaluated values, the network achieved consistently low approximation errors, demonstrating its ability to capture both the oscillatory behaviour and the amplitude decay.

For low damping values ($\beta < 0.3$), the model exhibited slightly higher variability between random seeds. This behaviour can be attributed to the fact that the system remains closer to the undamped oscillator regime, where the network must accurately preserve long-lasting oscillations throughout the entire prediction interval. For example, at ($\beta = 0.1$), the average MSE remained below 10^{-4} , although some individual seeds produced larger errors.

As the damping coefficient increased, the model generally maintained stable performance. Between ($\beta = 0.4$) and ($\beta = 1$), the MSE remained consistently in the range of 10^{-5} to 10^{-4} , while the maximum prediction error stayed below 4×10^{-2} for almost all experiments. The lowest individual error was obtained for ($\beta = 0.8$) with seed 2026, achieving an MSE of 3.98×10^{-6} and a maximum error of 3.9×10^{-3} .

The influence of random initialisation was also limited. Although some configurations showed small variations between seeds, the overall trend remained unchanged, indicating that the learned representation was not strongly dependent on the initial network weights.

Furthermore, a clear decrease in prediction error can be observed as β increases. Although the network already achieves accurate predictions for weakly damped systems, the average error progressively decreases with larger damping coefficients, reaching its lowest values as β approaches 1.

This behaviour can be explained by the physical characteristics of the

system. For small damping coefficients, the oscillator preserves its energy for longer periods, producing sustained oscillations that require the network to accurately predict the phase evolution over the entire simulation interval. Small deviations in the estimated frequency or phase accumulate over time, increasing the final prediction error.

In contrast, larger damping coefficients reduce the oscillation amplitude more rapidly, leading to smoother trajectories with lower long-term variability. Therefore, the neural network faces a simpler prediction task, as small deviations have a reduced impact on the final trajectory. This explains the observed reduction in MSE, MAE, and maximum error as the damping coefficient increases.

These results demonstrate that the neural network can successfully approximate damped oscillatory systems across a wide range of dissipation regimes. Compared with the harmonic oscillator, the additional exponential decay introduced by the damping term increases the complexity of the trajectory; however, the model maintains high prediction accuracy and stable generalisation performance.

3.3 Forced Harmonic Oscillator

The forced harmonic oscillator extends the damped harmonic oscillator by introducing a periodic external force. This additional term increases the complexity of the system, as the resulting motion depends on the interaction between the natural oscillation and the external excitation.

To evaluate this physical system, 65 experiments were performed, using 13 different values of the forcing angular frequency (Ω) and five different random seeds for each configuration. The reported metrics correspond to the average performance of the network for each value of Ω , while the standard deviation was used to evaluate the stability of the training process.

Additionally, the coefficient of determination (R^2) was included as an evaluation metric, defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (12)$$

where y_i represents the reference value, $\hat{y}_i$ represents the neural network prediction, $\bar{y}$ is the mean of the reference values, and n is the number of samples.

Figure 2 shows the results for both MSE and (R^2). Each value of (Ω) includes its corresponding mean error and standard deviation. It can be observed that for forcing frequencies below approximately ($\Omega = 2.5$), the

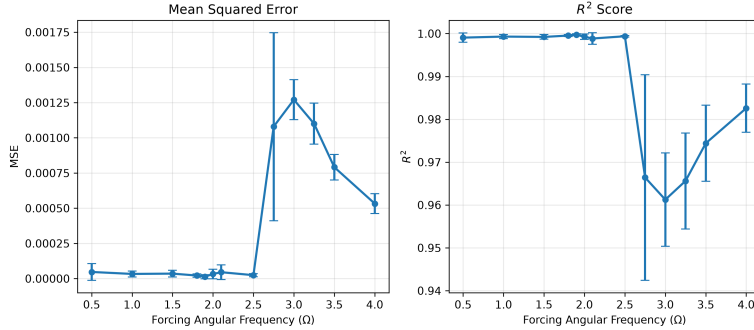


Figure 2: Prediction performance of the forced harmonic oscillator for different forcing angular frequencies. The figure shows the mean squared error (MSE) and coefficient of determination (R^2) obtained for each forcing frequency, including the variability across five independent training runs.

network achieved low approximation errors, whereas for higher frequencies the model began to show reduced prediction accuracy.

For example, for ($\Omega < 2.5$), the highest average MSE was obtained at ($\Omega = 0.5$), with a value of 4.73×10^{-5} . Throughout this range, the MSE remained consistently close to zero, indicating stable prediction performance. The lowest individual error was obtained for ($\Omega = 1.9$) with seed 57, achieving an MSE of 2.03×10^{-6} .

However, when ($\Omega > 2.5$), the network showed a decrease in prediction consistency. The lowest error in this regime, obtained at ($\Omega = 4.0$), was approximately eleven times larger than the highest error observed for ($\Omega < 2.5$). The largest error occurred at ($\Omega = 3.0$), with an MSE of 1.2×10^{-3} . Unlike the lower-frequency regime, where the error remained stable, the prediction accuracy fluctuated as the forcing frequency increased.

Figure 3 shows the temporal evolution of the prediction error for the experiment with ($\Omega = 3.0$) and seed 42. It can be observed that, after approximately ($t = 3$), the neural network prediction deviates significantly from the reference trajectory, converging towards values close to zero.

The degradation in prediction accuracy observed for forcing frequencies above approximately ($\Omega = 2.5$) requires further analysis. Although the forced harmonic oscillator remains a deterministic system, the difficulty of the learning task is not determined exclusively by the complexity of the governing equation, but also by the characteristics of the generated trajectory and its suitability for approximation by the neural network.

The forced harmonic oscillator contains two distinct dynamical com-

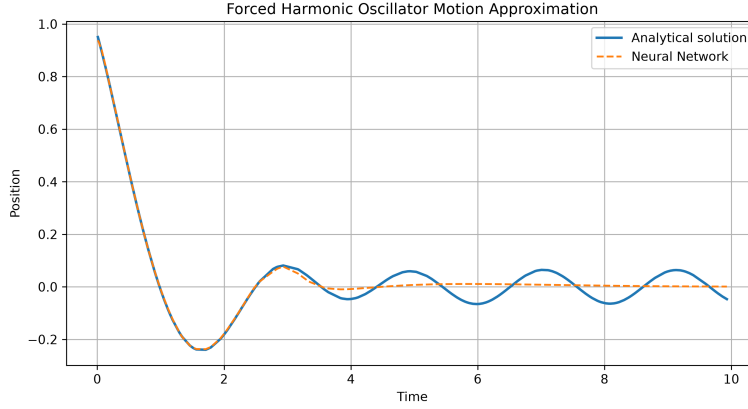


Figure 3: Numerical solution and neural network prediction for the forced harmonic oscillator with forcing angular frequency $\Omega = 3$ and random seed 42 over the simulation interval.

ponents: a transient response associated with the initial conditions and a stationary response [12] generated by the external excitation. The transient component decays exponentially according to $e^{-\beta t}$, becoming negligible after several seconds for the selected damping coefficient. However, during the initial stage of the trajectory, both components coexist, producing a signal composed of multiple oscillatory contributions.

For forcing frequencies close to the natural frequency of the oscillator, the resulting response remains relatively smooth [18], allowing the network to accurately approximate the trajectory. However, as the forcing frequency increases, the interaction between the natural dynamics and the external excitation produces more complex temporal patterns. The neural network must simultaneously learn the transient decay, the amplitude variation of the forced response, and the phase relationship between oscillations with different frequencies.

This effect becomes particularly relevant in the range ($\Omega = 2.75 - 3.5$), where the largest increase in prediction error is observed. In this regime, the trajectory combines relatively fast oscillations with a remaining transient contribution, creating a more challenging approximation problem. Since the model receives only the time variable as input, it does not have explicit information about the physical parameters and must infer the complete temporal relationship directly from the generated data.

Another possible contribution is related to the representational limitations of the multilayer perceptron architecture. Neural networks generally require

more capacity to approximate high-frequency components compared with smoother, lower-frequency functions [15]. Therefore, as the forcing frequency increases, the network may require additional complexity to represent the oscillatory structure with the same level of accuracy. The increase in error does not indicate physical instability of the oscillator, but rather that the selected architecture reaches a more challenging region of the function space.

Interestingly, the error does not increase indefinitely. For the highest tested frequency ($\Omega = 4.0$), the prediction accuracy improves compared with the intermediate-frequency regime. This suggests that the observed degradation is not caused solely by increasing oscillation frequency. Instead, it is likely related to the specific interaction between the transient and forced components within this frequency range. At larger frequencies, the transient contribution becomes relatively less significant compared with the forced oscillation, resulting in a simpler and more regular trajectory that can again be approximated more effectively by the neural network.

Further experiments with different architectures or longer simulation intervals would be required to isolate the contribution of each factor.

3.4 Duffing Oscillator

The Duffing oscillator represents the first nonlinear dynamical system investigated in this work. Unlike the previous oscillators, where the restoring force is proportional to the displacement, the Duffing equation introduces a nonlinear term that modifies the system response. This introduces additional complexity to the dynamics and provides a more challenging scenario for neural network prediction.

The system is defined by Equation 8, where the cubic term introduces nonlinear behaviour [17] and allows the system to exhibit complex phenomena such as amplitude-dependent oscillations and sensitivity to parameter variations [17].

The neural network architecture and training procedure were kept identical to those used in the previous experiments. The objective of this experiment was to evaluate whether the model, previously validated on linear systems, could successfully approximate nonlinear dynamics without requiring modifications to its structure.

The Duffing oscillator was evaluated through five independent parameter sweeps, one for each selected system parameter. For each sweep, five different parameter values were tested using five different random seeds, resulting in 25 training experiments per parameter. Figure 4 presents the mean MSE obtained for each explored parameter value, together with the variability

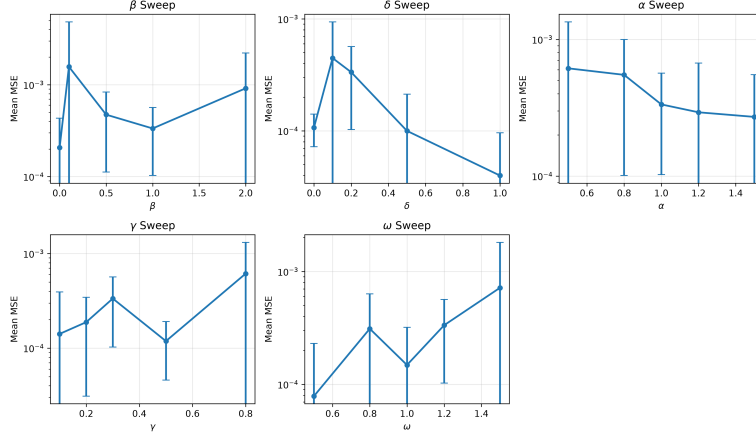


Figure 4: Mean squared error (MSE) of the Duffing oscillator for different parameter variations. The five subplots correspond to independent sweeps of the damping coefficient (δ), linear stiffness (α), nonlinear stiffness (β), forcing amplitude (γ), and forcing frequency (Ω), with results averaged over five random seeds.

caused by different training initialisations.

Unlike the previous experiments, no clear monotonic relationship was observed between the Duffing parameters and the prediction error. Although certain parameter values produced larger errors in individual executions, these variations were not consistent across different random seeds, suggesting that the observed differences are mainly related to training variability rather than a systematic increase in the difficulty of the learning task.

This behaviour indicates that, within the explored parameter range, modifying the Duffing parameters alone does not significantly increase the complexity of the prediction problem. Although the cubic restoring force introduces nonlinear dynamics and changes the system response, the resulting trajectories remain sufficiently smooth and regular for the neural network to approximate them accurately.

Therefore, the presence of nonlinear terms alone does not necessarily imply a significant degradation in predictive performance. Instead, these results suggest that the effective complexity of the generated trajectory is a more relevant factor than the mere presence of nonlinear terms in the governing equation.

3.5 Simple Pendulum

The simple pendulum represents a nonlinear physical system widely used as a benchmark for dynamical modelling. Unlike the harmonic oscillator, where the restoring force is proportional to the displacement, the restoring force of a pendulum depends on the sine of the angular position, resulting in nonlinear oscillatory behaviour for large amplitudes. The system is described by Equation 9, where g represents the gravitational acceleration and L the pendulum length. For small angular displacements, the approximation ($\sin(\theta) \approx \theta$) can be applied [18], reducing the system to a linear harmonic oscillator:

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad (13)$$

To evaluate the behaviour of the neural network on this physical system, five different parameter sweeps were performed. In the first sweep, the simulation duration was used as the variable parameter, with five experiments performed using the same random seed.

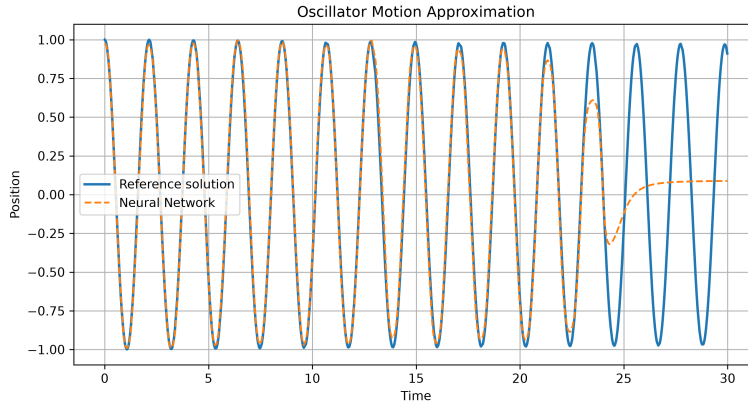


Figure 5: Numerical solution and neural network prediction for the simple pendulum over a 30-second simulation interval. The experiment was performed using random seed 42 and illustrates the long-term prediction behaviour of the model.

Figure 5 shows the experiment with a duration of 30 seconds and seed 42. The network accurately predicts the pendulum angle during most of the simulation interval. However, at approximately ($t = 24$) seconds, the prediction begins to deviate significantly, and the network loses the ability to reproduce the oscillatory behaviour accurately. Similar behaviour was observed for the 20-second experiment, while simulations with shorter durations maintained accurate predictions. Therefore, a duration of 10 seconds was selected for

the remaining sweeps in order to avoid long-term error accumulation while still providing sufficient trajectory information.

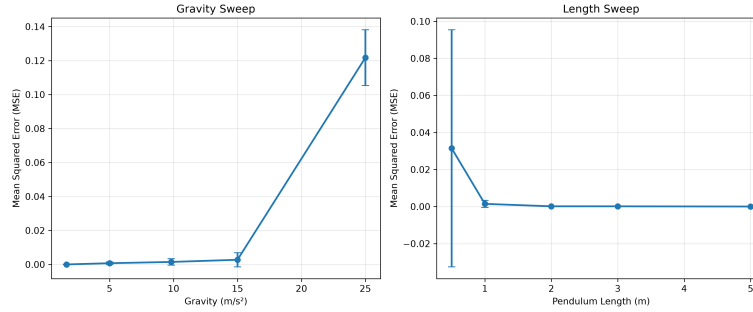


Figure 6: Mean squared error (MSE) of the simple pendulum for variations in gravitational acceleration (g) and pendulum length (L). The left subplot corresponds to the gravity sweep, while the right subplot corresponds to the length sweep. Results are averaged over five random seeds.

The next sweep investigated the effect of gravitational acceleration. This experiment contained five different gravity values and five random seeds per value, resulting in a total of 25 experiments. Figure 6 (left) shows the average MSE obtained for each gravitational acceleration.

For ($g < 15$), the MSE gradually increases but remains relatively low and consistent. However, for ($g = 25$), the neural network fails to approximate the dynamics accurately, reaching an average MSE of 0.12.

This behaviour can be explained by the dependence of the pendulum natural frequency on gravity. For small oscillations, the natural frequency is given by:

$$\omega_0 = \sqrt{\frac{g}{L}} \quad (14)$$

Therefore, increasing the gravitational acceleration increases the oscillation frequency. As observed in previous experiments with higher-frequency systems, the neural network shows greater difficulty representing faster temporal variations.

The length sweep followed the same structure, containing five different pendulum lengths with five random seeds each. Figure 6 (right) presents the average MSE for each length value. A behaviour similar to the gravity sweep can be observed, but in the opposite direction, since decreasing the pendulum length increases the natural frequency according to Equation 14.

The initial angle sweep also consisted of 25 experiments, with five initial angles evaluated using five random seeds each. Figure 7 shows the average

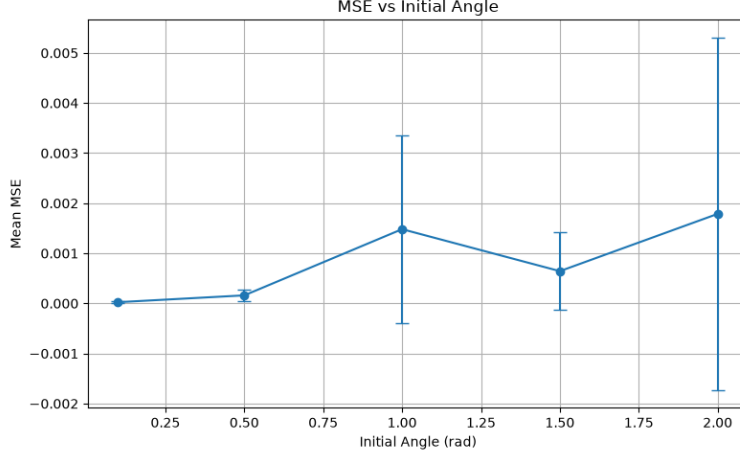


Figure 7: Prediction error of the simple pendulum for different initial angular displacements (θ_0). The figure shows the mean squared error (MSE) obtained for each value of the initial angle, averaged over five random seeds.

MSE for each initial angle.

The results show no clear monotonic relationship between the initial angle and prediction error. If the difficulty of the learning problem depended only on the degree of nonlinearity, a continuous increase in MSE would be expected as the initial angle increases, since larger angles produce a stronger deviation from the small-angle approximation. However, although the lowest angles present slightly lower errors, no significant increase in error is observed for larger initial angles.

These results suggest that nonlinearity alone is not sufficient to explain prediction error. Although increasing the initial angle makes the governing equation depart further from the linear approximation, the resulting trajectories remain smooth and periodic within the simulated time interval. Consequently, the multilayer perceptron remains capable of accurately approximating the solution.

The final sweep investigated the influence of the initial angular velocity. This experiment followed the same methodology as the previous sweeps, with five different initial velocities and five random seeds for each configuration. Figure 8 shows the average MSE for each initial angular velocity.

Increasing the initial angular velocity produces higher energy, larger oscillation amplitudes, and greater exploration of the nonlinear regime. Therefore, an initial hypothesis would be that larger initial velocities should

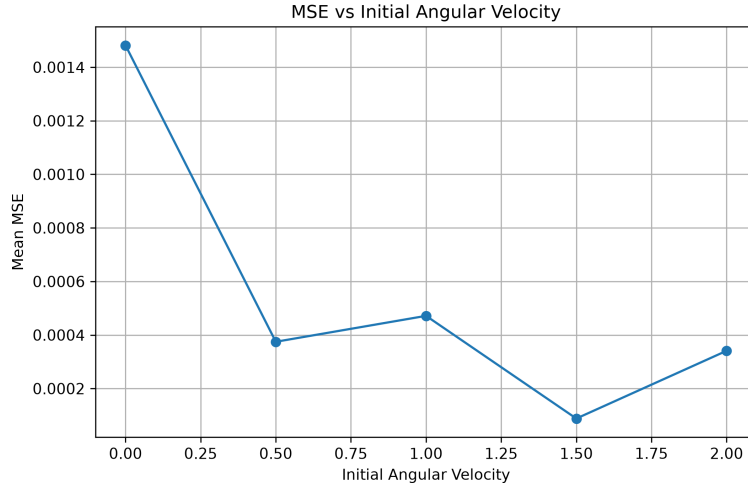


Figure 8: Mean squared error (MSE) of the simple pendulum for different initial angular velocities (ω_0). Results are averaged over five random seeds.

result in higher prediction errors.

However, similarly to the initial angle sweep, no direct relationship between velocity and error was observed. The highest error occurred for ($\omega_0 = 0$), with an MSE of 1.48×10^{-3} , whereas ($\omega_0 = 1.5$) achieved an MSE of 8.8×10^{-5} . Therefore, increasing the initial angular velocity does not necessarily make the prediction problem more difficult.

A slight overall decrease in prediction error can be observed as the initial angular velocity increases, although this trend is not strictly monotonic. Intermediate values exhibit small fluctuations, suggesting that the effect of initial angular velocity on prediction accuracy is relatively weak compared with parameters such as simulation duration or gravitational acceleration.

Overall, the simple pendulum experiments show that prediction accuracy is influenced more strongly by the temporal characteristics of the generated trajectory than by the mere presence of nonlinear terms in the governing equation. Parameters that increase oscillation frequency or extend the prediction horizon have a much greater impact on performance than those that only increase the degree of nonlinearity.

3.6 Double Pendulum

The double pendulum represents the most complex dynamical system analysed in this study. Unlike the simple pendulum, it consists of two coupled

pendulum arms, resulting in a system with two degrees of freedom and strongly nonlinear interactions between both components. The governing equations contain coupled nonlinear terms, making the resulting trajectories significantly more complex than those observed in previous systems.

For certain initial conditions, the double pendulum can exhibit chaotic behaviour [17], where small variations in the initial state may lead to substantially different trajectories over time [13]. This sensitivity to initial conditions represents an additional challenge for neural network-based modelling, as the objective is not only to approximate a given trajectory but also to generalise to unseen dynamical states.

To evaluate the performance of the neural network in this system, two types of experiments were performed. First, four parameter sweeps were conducted by varying the physical parameters of the system: the two pendulum lengths and the two initial angular positions. These experiments analyse how changes in the system configuration affect the prediction accuracy. Second, additional experiments were carried out to investigate the influence of sensitivity to initial conditions. In these experiments, small perturbations were introduced in the initial angle of the first pendulum to evaluate the ability of the model to generalise beyond the trajectory used during training.

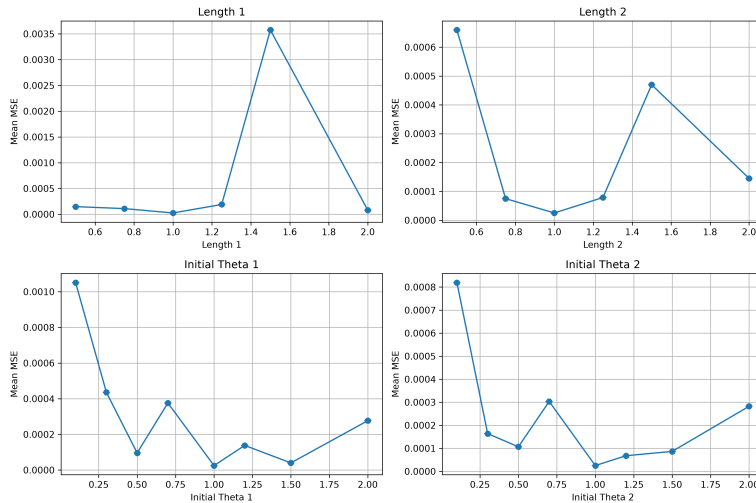


Figure 9: Mean squared error (MSE) of the double pendulum for different physical parameter variations. The figure shows the prediction error obtained by varying the initial angular positions (θ_1 , θ_2) and pendulum lengths (L_1 , L_2).

For the double pendulum system, four parameter sweeps were performed

by independently varying the two pendulum lengths and the two initial angular positions. Each sweep consisted of six different parameter values evaluated using a single random seed. Therefore, unlike previous experiments, the reported MSE values do not represent averages over multiple training runs, but the result obtained for each individual configuration. The objective of these experiments was to analyse how changes in the physical parameters affect the ability of the neural network to approximate the coupled nonlinear dynamics of the system.

The first two experiments investigated the influence of the pendulum lengths. Figure 9 shows the MSE obtained for the L_1 and L_2 sweeps. In both cases, no clear monotonic relationship between the length and the prediction error can be observed. Interestingly, the lowest error in both experiments corresponds to the reference configuration, where both lengths are equal to 1. For this case, the network achieved an MSE of 2.52×10^{-5} , indicating that this configuration generates a trajectory that is particularly suitable for approximation.

For the L_1 sweep, most configurations maintained low errors, with MSE values below 2×10^{-4} . However, when ($L_1 = 1.5$), the error increased significantly, reaching an MSE of 3.58×10^{-3} . This result shows that increasing the length of the first pendulum does not directly translate into higher prediction error. Instead, specific length configurations can generate more challenging dynamics for the neural network.

A possible explanation is that modifying L_1 changes the coupling between the two pendulums. Unlike the simple pendulum, where the length mainly determines the oscillation frequency, the double pendulum contains two interacting degrees of freedom. Changing the first pendulum length affects the energy transfer between both pendulums and can modify the resulting trajectory complexity. Therefore, certain configurations may produce less regular temporal patterns that are harder for the neural network to approximate.

The L_2 sweep showed a similar but less pronounced behaviour. The errors remained low for most configurations, although the smallest length value, ($L_2 = 0.5$), produced the highest MSE of 6.59×10^{-4} . Again, the reference configuration ($L_2 = 1$) achieved the lowest error. Since the second pendulum is influenced by the motion of the first one but does not directly control the entire system dynamics, variations in L_2 appear to have a smaller impact on the prediction difficulty.

The influence of the initial conditions was analysed through the θ_1 and θ_2 sweeps. As in the length experiments, the lowest error was obtained for the reference configuration, with ($\theta_1 = \theta_2 = 1$) rad resulting in an MSE of

2.52×10^{-5} . However, no monotonic relationship between the initial angle and the prediction error was observed.

For the θ_1 sweep, the largest error was obtained for an initial angle of 0.1 rad, with an MSE of 1.05×10^{-3} , while several larger angles achieved lower errors. Similarly, the θ_2 sweep showed variations between different initial angles, but without a clear tendency for the error to increase as the initial displacement increased.

This behaviour differs from what would be expected if the difficulty depended only on the degree of nonlinearity. For small angles, the simple pendulum approximation

$$\sin(\theta) \approx \theta \quad (15)$$

is valid, resulting in a system closer to a linear oscillator. For larger angles, this approximation becomes invalid and nonlinear effects become more significant. However, the results indicate that increasing nonlinearity alone is not enough to substantially degrade the performance of the neural network.

Overall, the four parameter sweeps show that the difficulty of learning double pendulum dynamics is not determined by a single physical parameter. Instead, the prediction error depends on the complexity of the resulting trajectory and on the interaction between the two coupled degrees of freedom. Most configurations remained within the approximation capabilities of the neural network, while specific combinations of parameters produced more complex dynamics and increased prediction errors. These results suggest that trajectory complexity is a more relevant factor than the magnitude of an individual physical parameter when evaluating the learning difficulty of nonlinear dynamical systems.

After analysing the influence of the physical parameters of the double pendulum, an additional experiment was performed to study one of the main characteristics of chaotic systems: sensitivity to initial conditions.

Unlike the previous sweeps, the objective of this experiment was not to determine how a parameter affects the prediction error of the neural network. Since each trajectory was trained and evaluated independently, similar prediction errors are expected for different initial conditions. Instead, this experiment aimed to analyse whether small variations in the initial state lead to significantly different system evolutions. All physical parameters were kept constant:

$$m_1 = m_2 = 1, \quad L_1 = L_2 = 1, \quad g = 9.81 \quad (16)$$

with initial conditions:

$$\theta_2 = 1.0, \quad \omega_1 = \omega_2 = 0 \quad (17)$$

The initial angle of the first pendulum was modified as:

$$\theta_1 = 1.0, 1.0001, 1.001, 1.01, 1.1 \quad (18)$$

These values represent increasingly larger perturbations around the same initial configuration. Each case was simulated and approximated by an independent neural network with the same architecture and training parameters.

Initial Angle	MSE	R^2
1.0	0.001191	0.997872
1.0001	0.000508	0.999064
1.001	0.000684	0.998787
1.01	0.000429	0.999253
1.1	0.002033	0.996512

Table 2: Prediction performance of the double pendulum for different initial values of θ_1 . Each row corresponds to an independent experiment performed with a single random seed.

The obtained prediction metrics are shown in Table 2. All configurations achieved similar performance, with R^2 values above 0.996. No clear relationship between the initial angle perturbation and the prediction error was observed. This result is expected, as each neural network learns its corresponding trajectory independently. Therefore, the differences in MSE do not represent the divergence between trajectories, but rather small variations in the approximation process.

To analyse the chaotic behaviour of the system, the difference between trajectories generated from different initial conditions was calculated:

$$|\Delta\theta_1(t)| = |\theta_{1,a}(t) - \theta_{1,b}(t)| \quad (19)$$

Figure 10 shows the evolution of this angular difference over time for the tested initial conditions. At the beginning of the simulation, the trajectories remain almost identical due to their extremely similar initial states. However, as time increases, the difference between them grows significantly. This divergence illustrates the sensitivity to initial conditions characteristic of chaotic systems [13]. Even small perturbations that are negligible at the

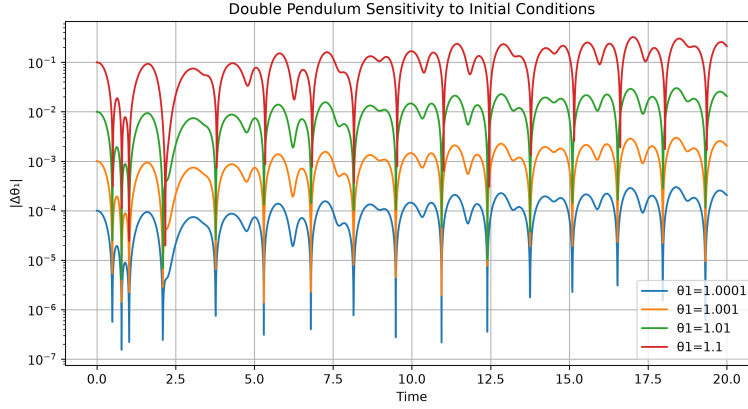


Figure 10: Evolution of the absolute angular difference $|\Delta\theta_1|$ for four double pendulum trajectories with slightly different initial conditions.

beginning can lead to substantially different future states. Consequently, two trajectories generated from initial conditions such as $(\theta_1 = 1.0)$ and $(\theta_1 = 1.0001)$ may evolve into noticeably different motions despite starting from almost identical configurations. This observation is important for neural network modelling of chaotic systems. A network trained on a single trajectory may accurately reproduce that specific evolution, but this does not necessarily imply that it has learned the underlying dynamics of the system. The strong dependence on the initial state suggests that generalizing to nearby but unseen trajectories represents a more challenging task. For this reason, the following experiment evaluates the ability of the neural network to predict trajectories generated from different initial conditions after being trained only on a single reference trajectory.

Following the previous analysis of sensitivity to initial conditions, a final experiment was performed to evaluate the ability of the neural network to generalize between nearby chaotic trajectories.

In contrast with the previous experiment, where each initial condition was trained and evaluated independently, in this case the neural network was trained using only a single trajectory with:

$$\theta_1 = 1.0 \quad (20)$$

while the evaluation was performed on trajectories generated from different initial conditions:

$$\theta_1 = 1.001, 1.01, 1.1 \quad (21)$$

All remaining physical parameters were kept constant.

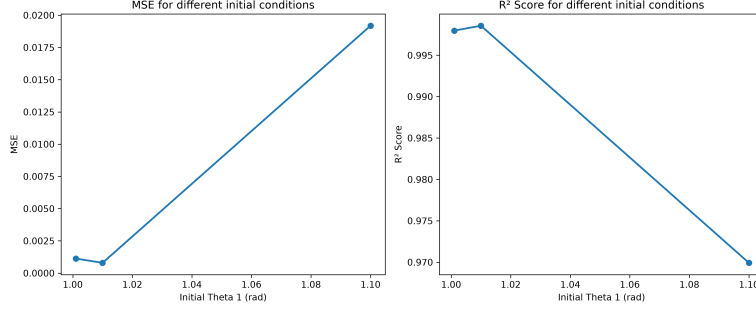


Figure 11: Generalization performance of the neural network trained with an initial angle of $\theta_1 = 1.0$. The left panel shows the mean squared error (MSE), while the right panel shows the coefficient of determination (R^2), evaluated for different initial values of θ_1 .

The results are shown in Figure 11. For small perturbations of the initial condition, the network maintains high prediction accuracy. The cases ($\theta_1 = 1.001$) and ($\theta_1 = 1.01$) achieve R^2 values of 0.998 and 0.999 respectively, indicating that the predicted trajectories remain close to the reference solution.

However, when the initial angle is changed to ($\theta_1 = 1.1$), the prediction error increases significantly, reaching an MSE of 1.92×10^{-2} and reducing the R^2 score to 0.9699. This degradation occurs because the evaluated trajectory progressively diverges from the trajectory used during training due to the sensitivity to initial conditions of the double pendulum.

These results highlight an important limitation of neural networks applied to chaotic systems. Although the model can accurately approximate a specific trajectory, learning one solution does not guarantee accurate prediction of other nearby trajectories. The chaotic nature of the system causes small differences in the initial state to produce different future evolutions, limiting the generalization capability of the model.

Overall, the double pendulum experiments show that neural networks can accurately approximate individual nonlinear trajectories. However, the results from the chaos test demonstrate a fundamental limitation: accurate prediction requires the model to be trained on the specific dynamical regime it is expected to reproduce. Small differences in initial conditions can lead to significant prediction errors due to the chaotic nature of the system.

4 Discussion

4.1 General performance of neural networks for dynamical systems

The results obtained throughout this work demonstrate that multilayer perceptrons (MLPs) are capable of accurately approximating a wide range of deterministic dynamical systems [11, 2], from simple linear oscillators to highly nonlinear systems. However, the experiments also show that prediction accuracy is not determined exclusively by the presence of nonlinear terms in the governing equations. Instead, the difficulty of learning a dynamical trajectory depends mainly on the characteristics of the generated solution, including its temporal complexity, oscillation frequency, and sensitivity to initial conditions.

The harmonic oscillator represented the simplest case studied, where the solution follows a regular periodic behaviour described by a sinusoidal function. As expected, the neural network achieved very low prediction errors, showing that a fixed-capacity MLP can accurately approximate smooth and repetitive trajectories. The introduction of damping and external forcing increased the complexity of the system by adding transient dynamics and additional oscillatory components. Nevertheless, the network maintained high accuracy for most explored parameter ranges, indicating that moderate increases in dynamical complexity do not necessarily prevent successful learning.

The Duffing oscillator further supports this conclusion. Despite being governed by a nonlinear differential equation containing a cubic restoring term [17], the neural network was able to reproduce the system dynamics with high precision for the investigated parameter values. The absence of a clear relationship between the nonlinear parameters and prediction error suggests that nonlinearity itself is not the main factor limiting neural network performance. Instead, the complexity of the resulting trajectory appears to be more relevant than the mathematical form of the governing equation. A nonlinear system producing smooth and regular trajectories can remain easier to approximate than a system with simpler equations but more complex temporal behaviour.

This observation is also supported by the simple pendulum experiments. Although the pendulum is a nonlinear system due to the presence of the sine restoring term, the neural network maintained accurate predictions for most tested configurations. The largest errors were associated mainly with parameters that increased the characteristic frequency of the motion, such

as higher gravitational acceleration or shorter lengths, suggesting that faster temporal variations represent a more significant challenge than nonlinearity alone.

The double pendulum provides an additional perspective on the capabilities and limitations of neural networks for dynamical modelling. Despite its chaotic nature, the network was still able to accurately approximate individual trajectories, as shown by the parameter sweeps performed over physical parameters and initial conditions. However, the chaos experiments revealed a different limitation: the difficulty of generalizing between trajectories generated from different initial conditions.

In chaotic systems, small variations in the initial state can lead to significantly different future trajectories due to sensitive dependence on initial conditions [13, 17]. Therefore, the increase in prediction error observed when evaluating the model on slightly different initial conditions does not indicate that the network cannot learn the double pendulum dynamics. Instead, it demonstrates that a model trained using only time as input,

$$t \rightarrow x(t), \tag{22}$$

learns the specific trajectory associated with the training conditions rather than the complete physical system. Consequently, when evaluated on a different dynamical realization, the model is attempting to reproduce a trajectory that it was not trained to represent.

Overall, these experiments indicate that the applicability of neural networks for dynamical systems is determined not simply by whether the system is linear or nonlinear, but by the properties of the resulting dynamics. Temporal complexity, oscillation frequency, and sensitivity to initial conditions play a more significant role in determining prediction performance. These results highlight both the potential of neural networks as approximators of physical trajectories and the importance of incorporating additional physical information when modelling complex dynamical systems.

4.2 Influence of temporal frequency and trajectory complexity

One of the main patterns observed throughout the experiments is that prediction accuracy is strongly influenced by the temporal characteristics of the trajectory. In particular, systems that exhibit faster oscillations or more rapidly changing dynamics tend to become more challenging for a fixed-capacity neural network [5], even when their governing equations remain relatively simple.

This behaviour was first observed in the forced harmonic oscillator experiments. While the network achieved very low prediction errors for most forcing frequencies, the performance degraded when the excitation frequency increased beyond a certain range. Increasing the forcing frequency introduces faster oscillatory components into the solution, reducing the temporal scale over which the network must approximate changes in the signal. Since the model receives only time as input, it must represent increasingly rapid variations using the same fixed architecture, which can limit its approximation capability [19].

A similar effect was observed in the simple pendulum experiments. For small oscillations, the natural frequency of the pendulum can be approximated as in Equation 14, where g is the gravitational acceleration and L is the pendulum length. Therefore, increasing g or decreasing L increases the characteristic frequency of the system. The gravity and length sweeps showed that configurations with larger frequencies generally produced higher prediction errors. For example, increasing the gravitational acceleration beyond the explored normal range caused a significant increase in error, while shorter pendulum lengths also resulted in less accurate predictions. These results suggest that the network is more sensitive to rapid temporal variations than to the physical origin of those variations.

This relationship can also explain the behaviour observed in the forced oscillator and pendulum experiments where increasing frequency did not always lead to a monotonic increase in error. The difficulty of approximation depends not only on the frequency value itself but also on the complete structure of the trajectory. Factors such as transient components, amplitude, and the interaction between different oscillatory modes can influence how complex the resulting signal is for the neural network to approximate.

The results obtained with the Duffing oscillator further reinforce this interpretation. Although the system contains nonlinear terms that modify the shape of the trajectory, the explored parameter variations did not produce a consistent increase in prediction error. This indicates that nonlinear behaviour alone does not necessarily create a difficult learning problem. Instead, the relevant factor is whether the nonlinearity generates complex temporal patterns that exceed the approximation capabilities of the model.

Therefore, the experiments suggest that the difficulty of learning dynamical systems is closely related to the temporal complexity of their solutions. A system with a nonlinear governing equation can remain easy to approximate if its trajectory is smooth and regular, whereas a system with faster or more complex temporal variations may become challenging even without strong nonlinear effects. This highlights the importance of analysing the

characteristics of the generated trajectory, rather than only the mathematical complexity of the underlying physical model, when evaluating neural network approaches for dynamical modelling

4.3 Limitations of purely data-driven dynamical modelling

Although the results demonstrate that neural networks can accurately approximate a wide range of dynamical systems, the experiments also reveal important limitations associated with purely data-driven approaches. The main limitation is that the model learns a mapping between the provided input and the observed trajectory, rather than explicitly learning the physical laws governing the system [16].

In this work, the neural network receives only time as input and predicts the corresponding system state, as described in Equation 22.

Therefore, the model is trained to approximate a specific solution generated under fixed physical parameters and initial conditions. While this approach is sufficient to reproduce individual trajectories with high accuracy, it does not provide the network with information about the underlying physical system, such as masses, lengths, gravitational acceleration, or initial conditions.

This limitation becomes particularly evident in the double pendulum chaos experiments. The network successfully approximates the trajectory used during training, but its performance decreases when evaluated on a different trajectory generated from slightly modified initial conditions. This behaviour is expected in chaotic systems, where small differences in the initial state can lead to large deviations over time. The observed error increase is therefore not caused by an inability of the network to represent the dynamics, but by the fact that the model was not trained to represent the complete family of possible solutions.

A similar limitation can be observed in the parameter sweeps performed throughout this work. When the network is trained and evaluated on the same physical configuration, it achieves very accurate predictions. However, the experiments were designed independently for each parameter value, meaning that each model only learned one specific dynamical regime. As a result, the trained networks cannot be expected to generalize automatically to unseen combinations of physical parameters.

This distinction is important when considering the application of neural networks to physical modelling. A model that accurately reproduces a single trajectory does not necessarily represent a general surrogate model of the physical system. For applications requiring prediction across different

operating conditions, the relevant physical parameters should be included as inputs, allowing the network to learn a broader representation of the system dynamics.

For example, instead of learning only the input-output relation described in Equation 22, a more general formulation would be:

$$(t, m, L, g, \theta_0, \omega_0) \rightarrow x(t), \quad (23)$$

where the physical parameters and initial conditions are provided explicitly. This approach would allow the network to learn the relationship between the governing parameters and the resulting trajectories, improving its ability to interpolate between different dynamical configurations [14].

Therefore, while the results confirm the capability of neural networks as trajectory approximators, they also highlight the importance of distinguishing between trajectory fitting and physical system identification. Incorporating physical information into the learning process represents a promising direction for improving the reliability and generalization of neural network-based dynamical models.

4.4 Future improvements and physics-informed approaches

The results obtained in this work demonstrate the potential of neural networks for approximating dynamical systems, but they also highlight the limitations of purely data-driven approaches. In particular, the experiments show that a model trained only on observed trajectories can achieve excellent accuracy for specific configurations while having limited generalization capabilities when the underlying dynamics change. Future improvements should therefore focus on incorporating additional physical information into the learning process.

One possible approach is the use of physics-informed neural networks (PINNs) [16], which incorporate the governing equations of the physical system directly into the training process. Unlike traditional neural networks, which minimize only the difference between predictions and observed data, PINNs include an additional physics-based loss term that penalizes violations of the differential equations describing the system.

The loss function can therefore be expressed as:

$$Loss = Loss_{\text{data}} + Loss_{\text{physics}} \quad (24)$$

where $Loss_{\text{data}}$ measures the difference between the predicted and actual trajectories, while $Loss_{\text{physics}}$ evaluates how well the prediction satisfies the underlying equations of motion. By combining experimental data with

physical constraints, the model is encouraged to learn the actual dynamics rather than only fitting the available trajectories.

Another possible improvement would be to include physical parameters and initial conditions as inputs to the neural network. In this work, each experiment required training a separate model for each set of parameters. A more general model could instead learn a mapping of the form:

$$(t, p_1, p_2, \dots, p_n) \rightarrow x(t), \quad (25)$$

where p_i represents relevant physical parameters such as mass, length, gravity, damping coefficient, or initial conditions. This would allow the network to learn a continuous representation of the system and potentially predict trajectories for previously unseen configurations.

Additionally, the architecture of the neural network could be adapted depending on the characteristics of the system being modelled. The multi-layer perceptron used in this work approximates the trajectory as a static function of time. However, dynamical systems naturally contain temporal dependencies, especially when predicting long-term evolution. Recurrent architectures, such as long short-term memory networks (LSTMs) or gated recurrent units (GRUs) [8, 4], could introduce memory into the model and improve the representation of sequential information. Another promising direction is the use of Neural Ordinary Differential Equations (Neural ODEs) [3], where the neural network directly learns the evolution rule of the system instead of approximating the final trajectory.

For chaotic systems, such as the double pendulum, perfect long-term prediction remains fundamentally limited due to sensitive dependence on initial conditions [13]. Even a physically accurate model cannot overcome the intrinsic unpredictability caused by chaotic divergence. However, incorporating the governing equations and physical parameters can improve short-term prediction accuracy and allow the model to reproduce the underlying dynamics more reliably.

Overall, the experiments suggest that the future of neural network-based dynamical modelling lies not in replacing physical knowledge with data, but in combining both approaches. Data-driven methods provide flexibility and approximation capabilities, while physical information provides constraints that improve generalization, interpretability, and reliability.

5 Conclusion

This work investigated the ability of a multilayer perceptron (MLP) to approximate the dynamics of several physical systems with increasing levels of complexity, ranging from the linear harmonic oscillator to the chaotic double pendulum. A common neural network architecture was trained using only time as input, allowing the influence of the underlying system dynamics on the prediction accuracy to be analysed independently of architectural changes.

The results show that the neural network successfully learned the behaviour of all considered systems. For the harmonic oscillator and the damped harmonic oscillator, prediction errors remained extremely low across all experiments, confirming that smooth and regular trajectories can be accurately approximated by a relatively simple feedforward network [5]. The forced harmonic oscillator introduced a more demanding learning task due to the interaction between the natural oscillation and the external forcing frequency. Although prediction accuracy remained high for most forcing frequencies, the experiments revealed that high-frequency oscillations produced a noticeable increase in prediction error, suggesting that rapidly varying trajectories are intrinsically more difficult for the network to approximate.

The Duffing oscillator demonstrated that the presence of nonlinear terms alone does not necessarily reduce predictive performance. Across the explored parameter ranges, variations in the nonlinear coefficients produced only minor changes in the approximation error. This indicates that the complexity of the generated trajectory is more relevant than the mathematical form of the governing equation itself.

The simple pendulum produced similar conclusions. Changes in gravity and pendulum length affected prediction accuracy primarily because they modified the oscillation frequency, while variations in the initial angle or initial angular velocity had comparatively little influence. Furthermore, the duration experiments showed that the network could accurately predict long trajectories only up to a certain time horizon, after which the accumulated approximation error became increasingly visible.

The double pendulum represented the most complex system considered in this work. Despite its nonlinear and chaotic behaviour, the neural network achieved high prediction accuracy when training and evaluation were performed under the same physical conditions. However, the chaos generalisation experiments demonstrated an important limitation. A network trained on one trajectory did not necessarily generalise well to trajectories generated from nearby initial conditions once the divergence between the trajectories

became sufficiently large. This behaviour reflects the intrinsic sensitivity to initial conditions of chaotic systems [13, 17] rather than an inability of the neural network to fit an individual trajectory.

Overall, the experiments suggest that the primary factor determining prediction difficulty is not whether a system is linear or nonlinear, but rather the complexity of the trajectory that must be approximated. Systems exhibiting higher oscillation frequencies, longer prediction horizons, or chaotic divergence generally require the network to approximate more rapidly varying functions [19], leading to increased prediction errors even when the underlying governing equations remain deterministic.

Several limitations should also be acknowledged. Only a single feed-forward MLP architecture was considered, and all experiments focused on deterministic simulations without measurement noise. Future work could investigate recurrent neural networks, transformer-based models, or neural operators capable of exploiting temporal dependencies more effectively [8, 4]. Additionally, extending the analysis to longer prediction horizons, more strongly chaotic regimes, and physics-informed neural networks could provide further insight [16] into the relationship between machine learning models and nonlinear dynamical systems.

In summary, this work demonstrates that relatively simple neural networks can accurately approximate a broad range of physical systems, including nonlinear and chaotic dynamics, provided that the trajectories remain sufficiently smooth and the prediction task stays within the interpolation regime. The results highlight both the potential and the limitations of purely data-driven approaches for modelling physical systems and provide a systematic comparison of how different dynamical properties influence neural network performance.

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